
Comparing Semantic and Logical Composition Using Latent Diffusion Models

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Abstract

This draft adapts Phase 1 of a broader research programme into a compact NeurIPS-style paper. The paper studies the mismatch between semantic composition from a single text prompt and logical composition from score-level product-of-experts guidance in latent diffusion models. Sections 3–6 focus on the Phase 1 problem formulation, the planned experimental structure, the expected trajectory-level and reachability trends, and the interpretation of those expected outcomes.

1 Introduction

Generative artificial intelligence has advanced rapidly in recent years. The strong performance and wide adoption of systems such as Stable Diffusion have increased interest in models that generate structured content from flexible conditioning signals [Rombach et al., 2022]. Generative modeling is now used in natural language processing, audio generation, image synthesis, super-resolution, medical imaging, and video generation [Brown et al., 2020, Borsos et al., 2022, Ramesh et al., 2021, Saharia et al., 2021, Peng et al., 2022, Ho et al., 2022]. Among these domains, text-to-image generation is a useful setting for analysis because it requires a model to map a linguistic description to a coherent visual scene while preserving the structure of the prompt. This makes it possible to study not only image quality, but also whether the model composes meaning reliably.

One appealing response is *compositional diffusion*: instead of training a new model for every conjunction, reuse pretrained components at inference time and combine their guidance signals directly [Du et al., 2020, Liu et al., 2022, Du and Kaelbling, 2024]. In latent diffusion models, this creates two natural ways to request what appears to be the same scene. The first is *semantic composition*, where a single joint prompt such as “a cat and a dog” is passed through the standard text-conditioning pipeline. The second is *logical composition*, where the model combines single-concept conditionals through a score-level product-of-experts (PoE) rule to implement conjunction explicitly.

These two routes are often treated as neighboring versions of the same task, but most prior work has focused on whether a composition rule improves prompt fidelity, reduces collapse, or stabilizes inference [Du et al., 2023, Du and Kaelbling, 2024, Dutta et al., 2026]. Much less is known about whether a monolithic prompt and a PoE composition actually target the same solution under shared noise, or about how any disagreement depends on the structural relation between the concepts being composed. As a result, current evaluations can show that a composed image looks better or worse without clarifying whether semantic prompting and logical composition are even operating on the same latent objective.

This paper studies that mismatch directly. We define the *composability gap* as the difference between the latent trajectories and endpoints reached by semantic composition and logical composition when both start from the same initial latent. The question matters because, if the two mechanisms separate under controlled conditions, then PoE guidance is not merely another way to phrase the same request. It may be steering sampling toward a different target than the one represented by the joint prompt in the model’s learned distribution.

Our central claim is that this gap is structural rather than uniform. Following the conditions identified by Bradley et al. [2025], semantic and logical composition should remain close when the desired joint configuration is already supported by the model or when the relevant factors are sufficiently separable. The gap should grow when those conditions weaken: first for partially overlapping pairs, and most strongly for *coherent collisions*, where concepts remain semantically plausible together but compete for the same identity, role, or spatial slot. This perspective leads to a four-part taxonomy of supported co-occurrence, approximate factorization, partial overlap, and coherent collision.

Figure 1 provides the intended intuition before the formal setup. Each panel fixes the initial noise and compares solo prompts, the monolithic joint prompt, and PoE composition for one representative pair from one taxonomy regime. The reader should be able to see the paper’s main claim at a glance: semantic and logical composition stay coupled in easier regimes, begin to fan out in intermediate regimes, and diverge most clearly when concept interaction becomes compositionally fragile.

Guided by this hypothesis, we compare monolithic prompting and PoE guidance under a shared-noise protocol, track their full latent trajectories and terminal gaps, and then ask a reachability question: if PoE reaches a particular endpoint, can that target be recovered by projecting the PoE image back into the model’s conditioning space and rerunning text-conditioned generation? This design separates a weak explanation—that the prompt was simply phrased differently—from a stronger one in which logical composition steers toward targets that standard semantic prompting does not reliably reproduce.

The result matters at two levels. Practically, it provides a way to evaluate when plug-and-play composition methods track the intended joint scene and when they instead induce a different generative objective. Scientifically, it clarifies what latent diffusion models encode as joint meaning through standard text conditioning versus what is imposed by inference-time score composition. The rest of the paper formalizes the semantic and logical composition rules, introduces the concept-pair taxonomy and shared-noise protocol, analyzes trajectory divergence, and studies reachability through SD-IPC reruns.

2 Related Work

Inference-time composition. Work on compositional generation has largely focused on how pre-trained experts can be combined at sampling time. Compositional energy-based models showed that conjunctions, disjunctions, and negations can be expressed through products of experts over concept-specific energies [Du et al., 2020], and latent-space energy-based models extended this idea to controllable generation without retraining the base generator [Nie et al., 2021]. In diffusion models, Liu et al. [2022] provide the most direct precursor to our setting by showing that classifier-free guidance terms from a pretrained text-to-image model can be combined to implement logical operators such as AND and NOT. Subsequent work broadened this family rather than resolving when it should work: Du et al. [2023] and Du and Kaelbling [2024] introduced MCMC-inspired samplers, energy-based parameterizations, and dynamic guidance schedules to stabilize composition; Skreta et al. [2024] proposed SuperDiff, which combines diffusion models through an Itô density estimator; Zhang et al. [2025] generalized product-of-experts composition to heterogeneous visual generators; and Dutta et al. [2026] added corrective sampling to steer away from unstable mixed modes. In a different sense of the term, CoDi studies any-to-any multimodal generation via composable diffusion [Tang et al., 2023], but its goal is cross-modal generation rather than logical conjunction inside one text-to-image diffusion model.

Theory and failure mechanisms. More recent work has shifted from proposing operators to asking why composition succeeds or fails. Bradley et al. [2025] provide the main theoretical reference point for this paper by defining projective composition and showing that Bayes-style score composition is reliable only when the desired joint configuration has compatible support or when the relevant factors are sufficiently separable in some feature space. When these conditions fail, the combined score can become incoherent, producing collapse, hybridization, or dominance rather than a faithful joint sample. Mitigation papers such as CO3 [Dutta et al., 2026] target one symptom of this problem by steering sampling away from mode collisions, but they do not explain how failure should vary across different types of concept pairs. Our work instead operationalizes these structural conditions into a concept-pair taxonomy and tests whether the gap between semantic prompting and logical

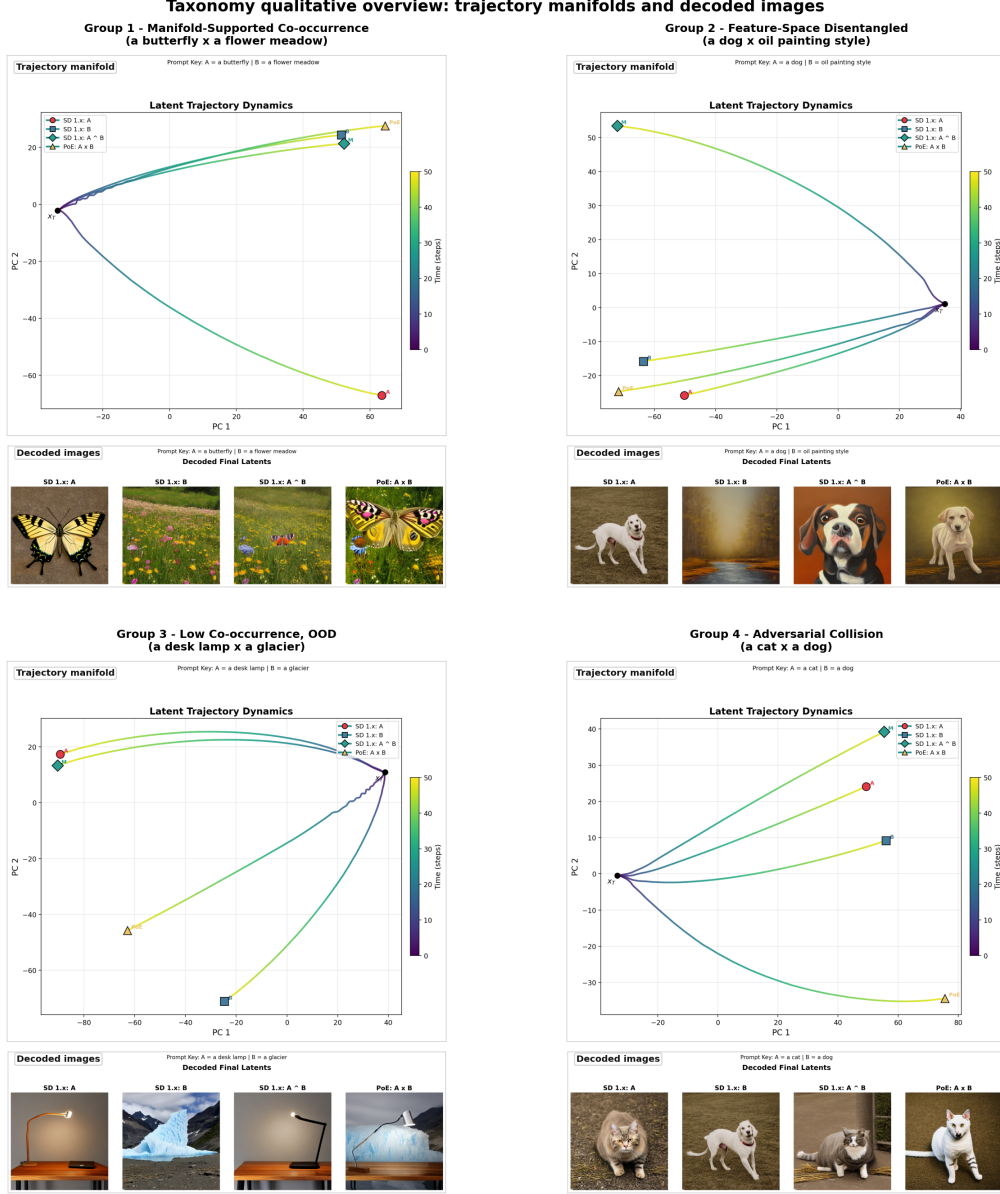


Figure 1: Representative shared-noise taxonomy panels for one pair from each Phase 1 group. In every panel, the upper axis shows a two-dimensional MDS projection of the full denoising trajectories for solo A , solo B , the monolithic prompt $A \wedge B$, and logical composition by PoE $A \times B$, all starting from the same initial latent x_T . The lower strip shows the decoded final images for the same four conditions in the same left-to-right order. The panels are illustrative rather than inferential: they give a qualitative view of when semantic and logical composition remain coupled and when they fan out under shared noise before the group-wise statistics are introduced.

product-of-experts composition follows a predictable ordering across co-occurring, disentangled, overlapping, and coherent-collision regimes.

Benchmarks and evaluation. A complementary line of work evaluates whether text-to-image models can satisfy compositional prompts in the first place. T2I-CompBench [Huang et al., 2023] and its expanded TPAMI follow-up T2I-CompBench++ [Huang et al., 2025] benchmark prompt-level compositionality across attributes, relations, numeracy, and complex scenes, showing that even strong generators remain brittle on open-world compositions. Those benchmarks are valuable for measur-

ing prompt fidelity, but they are not designed to compare semantic composition against inference-time logical composition under matched initial conditions, nor to explain why some concept pairings are systematically harder than others. Our evaluation therefore uses a theory-grounded concept-pair taxonomy, shared-noise latent trajectory analysis, and an SD-IPC-based reachability test [Ding et al., 2023] to ask a different question: not only whether a composed image looks correct, but whether logical score composition reaches latent targets that semantic prompting can reproduce, and whether that mismatch can be predicted from the relation between the constituent concepts. In that sense, prior work shows how to compose, stabilize, or benchmark composition, whereas this paper characterizes when semantic and logical composition align or diverge across concept regimes.

3 Problem Setup

Recent work on compositional generation has shown that pretrained generative models can be combined at inference time by composing conditional score functions rather than relying on a single joint prompt [Du et al., 2020, Du and Kaelbling, 2024, Liu et al., 2022]. In latent diffusion models, the simplest form of this idea is a Bayes-style composition rule:

$$\nabla_x \log p(x \mid c_1, c_2) \approx \nabla_x \log p(x \mid c_1) + \nabla_x \log p(x \mid c_2) - \nabla_x \log p(x), \quad (1)$$

which implicitly defines a product-of-experts (PoE) distribution

$$\hat{p}(x) \propto \frac{p(x \mid c_1)p(x \mid c_2)}{p(x)}. \quad (2)$$

This construction treats composition as logical conjunction implemented directly in score space.

Throughout the paper, *semantic composition* means generation from a single joint text prompt, such as “ c_1 and c_2 ,” while *logical composition* means score-level conjunction via the Bayes-style PoE rule above. The empirical object of interest is the *composability gap*: the discrepancy between the latent trajectories and endpoints reached by those two mechanisms under the same initial noise. If the model represents a compatible joint mode, or if the relevant factors are sufficiently separable, then this gap should remain small; when neither condition holds, logical composition may become unstable or degenerate [Bradley et al., 2025].

Phase 1 tests whether that gap follows a structured ordering across four concept-pair regimes: manifold-supported co-occurrence, feature-space factorization, weaker overlap without clean factorization, and coherent collision. The goal is not only to show that semantic and logical composition can differ, but to determine whether the size and timing of that difference are predictable from the relation between the constituent concepts.

4 Experiments

4.1 Evaluation Set and Concept Pair Taxonomy

Design Principle. The evaluation set is a controlled collection of concept pairs designed to test the structural conditions under which logical score composition should align with, or diverge from, semantic composition. Following Bradley et al. [2025], Bayes-style composition is expected to behave well under two broad conditions: either the joint distribution is supported in the training data, or the relevant conditionals approximately factorize in a suitable representation. The taxonomy is therefore constructed to span regimes in which these conditions hold, weaken, or fail strongly, so that the composability gap can be studied as a structured phenomenon rather than through isolated examples. Phase 1 uses four taxonomy groups, with six concept pairs per group and twenty-four shared-noise seeds per pair, yielding $6 \times 24 = 144$ pair-seed records per group and 576 total pair-seed records across the study.

Taxonomy Groups. Phase 1 uses four pair regimes, each defined by a different structural relationship between data support and factorization.

Group 1: Manifold-Supported Co-occurrence. These pairs correspond to concept combinations for which the model is expected to already represent a usable joint mode, such as camel–desert

or boat–ocean. Semantic composition and PoE should therefore remain close, since composition recovers an already supported configuration rather than requiring substantial extrapolation. This group serves as the positive control for data-supported composition.

Group 2: Feature-Space Factorization. These pairs need not co-occur frequently in the training distribution, but are chosen so that the concepts act on approximately distinct components, as in object–style combinations such as dog–watercolor or car–cubist painting. Here the expected agreement between semantic composition and PoE comes from structural separability in representation space rather than from direct empirical co-occurrence. This group serves as the positive control for representation-supported composition.

Group 3: Weak Support and Non-Factorized Structure. These pairs are selected to avoid both clear co-occurrence and clean factorization, creating an intermediate regime in which the composability gap should increase but remain heterogeneous. Within this group, we distinguish missing joint support, feature entanglement, and partial interference as three motivating failure modes for pair selection. Operationally, Group 3 is constructed with CLIP-proxy alignment heuristics to target combinations that are neither cleanly factorized nor maximally adversarial, so some pairs may still admit only partial or unstable solutions rather than complete failure.

Group 4: Coherent Collision. These pairs compete for the same semantic role, object identity, or spatial slot, as in cat–dog or cat–owl. In this regime, semantic composition and PoE are expected to diverge most strongly, often through hybridization, suppression, or collapse to a dominant mode, because the relevant degrees of freedom are not plausibly separable. This group is the strong-failure regime and is expected to show the largest and most systematic composability gap.

Expected Ordering and Analytic Role. Taken together, the taxonomy defines the expected ordering

$$G1 \approx G2 < G3 < G4,$$

with Groups 1–2 serving as the two success regimes, Group 3 capturing heterogeneous intermediate failure, and Group 4 capturing the strongest violations of the compositional assumptions. Phase 1 uses this taxonomy to test whether the composability gap follows that ordering at the level of pairwise latent trajectories and endpoints rather than only in isolated examples. The same taxonomy also separates failures driven primarily by missing data support from failures driven by representation-level entanglement or interference, which motivates the Phase 2 question of whether improving representation quality can selectively reduce the latter class of errors. Figure 1 gives the qualitative shared-noise grounding view before the group-wise quantitative analysis.

4.2 Hyperparameters and Shared-Noise Protocol

All Phase 1 runs use Stable Diffusion 1.4 as the primary latent diffusion model [Rombach et al., 2022]. For every concept pair and seed, all generation conditions begin from the same initial latent state x_T , so any difference between conditions is attributable to the composition rule rather than stochastic initialization. The core comparison is between semantic composition from a monolithic joint prompt and logical composition from PoE guidance.

The denoising schedule uses 50 steps throughout Phase 1. The same shared-noise protocol is applied to solo c_1 , solo c_2 , the monolithic prompt, and PoE, but group-level inference is performed only after aggregating repeated seeds within each concept pair. Concretely, the concept pair is the formal unit of inference, while the twenty-four seeds act as repeated measurements for estimating pair-level summaries.

Phase 1 also includes a reachability test using SD-IPC [Ding et al., 2023]. For each pair-seed record, the decoded PoE image is projected back into the Stable Diffusion conditioning space to obtain a continuous pseudo-prompt p^* . Stable Diffusion 1.4 is then rerun from the same initial noise using p^* , allowing the regenerated trajectory to be compared directly against the original PoE trajectory on the same shared-noise axis.

4.3 Expected Results: Latent Trajectory Dynamics

The primary trajectory metric is the per-step latent gap between semantic and logical composition:

$$d_t^{(p,s)} = \frac{1}{N} \left\| z_t^{\text{mono},(p,s)} - z_t^{\text{poe},(p,s)} \right\|_2^2, \quad (3)$$

where p indexes the concept pair, s indexes the shared-noise seed, and N is the number of latent elements. Its terminal value

$$d_T^{(p,s)} = d_t^{(p,s)}|_{t=T} \quad (4)$$

measures the final latent mismatch between the two composition mechanisms. Pair-level summaries are then formed by averaging over the 24 seeds for each concept pair before any group-wise comparison is made.

The expected result is an ordered taxonomy effect. Groups 1 and 2 should exhibit the smallest pair-level terminal gaps because these regimes correspond either to supported co-occurrence or to approximate feature factorization. Group 3 should show a broader and more variable range of pair-level gaps, reflecting partial structural incompatibility. Group 4 is expected to show the largest and most systematic separation, consistent with hybridization, dominance, or collapse when both concepts compete for the same semantic role.

The same ordering is expected to appear in the temporal profiles $\bar{d}_t^{(p)}$. The semantic-logical discrepancy may be detectable before decoding, with trajectories separating during denoising rather than only at the final image. Figure 1 provides the representative qualitative view of this fan-out geometry. Phase 1 specifically anticipates that divergence will become more pronounced in later low-noise steps, where the model must resolve fine-grained object identity and attribute details under conflicting compositional pressures.

4.4 Expected Results: Reachability Analysis

The reachability analysis asks whether the endpoint produced by logical composition can be recovered by returning to the model through language-conditioned generation. For each pair-seed record, the SD-IPC pseudo-prompt rerun produces a regenerated trajectory $\{z_t^{*,(p,s)}\}$, which is compared against the original PoE trajectory $\{z_t^{\text{poe},(p,s)}\}$ using

$$d_t^{*,(p,s)} = \frac{1}{N} \left\| z_t^{*,(p,s)} - z_t^{\text{poe},(p,s)} \right\|_2^2, \quad d_T^{*,(p,s)} = d_t^{*,(p,s)}|_{t=T}. \quad (5)$$

Figure 2 repeats the same representative taxonomy layout after adding the SD-IPC rerun from p^* . By keeping the pair selection and shared-noise seed fixed, the figure asks the closure question directly: does the projected conditioning return to the PoE basin, or does it remain closer to semantic composition?

Under the 50-step schedule, each residual curve is also summarized over three denoising phases:

$$\Delta_{\text{early}}^{*,(p,s)} = d_{17}^{*,(p,s)} - d_0^{*,(p,s)}, \quad (6)$$

$$\Delta_{\text{mid}}^{*,(p,s)} = d_{34}^{*,(p,s)} - d_{17}^{*,(p,s)}, \quad (7)$$

$$\Delta_{\text{late}}^{*,(p,s)} = d_{50}^{*,(p,s)} - d_{34}^{*,(p,s)}. \quad (8)$$

Image-level CLIP cosine similarity between the regenerated image and the original PoE image is reported as a secondary output-space measure.

The expected result is again taxonomy dependent. In Groups 1 and 2, where the original composability gap is predicted to be small, the p^* rerun should remain relatively close to the PoE basin and produce small terminal residuals. In Groups 3 and 4, the rerun is expected to drift away from the PoE trajectory, indicating that the logical target is not fully recoverable through SD-IPC-conditioned generation alone. Early residual growth would suggest failure to enter the PoE basin from the outset, whereas late residual growth would indicate only partial alignment followed by terminal breakdown.

At the group level, successful gap closure would appear as a left shift of the d_T^* distribution toward zero and weaker phase-wise residual accumulation than the original monolithic-to-PoE baseline.

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Figure 2: Representative taxonomy panels after adding the SD-IPC rerun from p^* . If SD-IPC closes the gap, its decoded images and trajectory manifolds should move toward the PoE outputs for the same shared-noise seed; persistent separation indicates that the logical target is not fully recoverable through text-conditioned rerunning alone.

The Phase 1 expectation, however, is that closure will be limited in the harder regimes, especially for coherent-collision pairs. In that case, reachability analysis would support the claim that PoE composition can steer sampling toward targets that a single recovered conditioning signal does not reproduce faithfully.

5 Discussion

If the expected ordering across taxonomy groups is observed, Phase 1 would support a structural interpretation of the composability gap. The key point would not be merely that monolithic prompting and PoE differ, but that the magnitude and timing of their divergence depend on the compatibility structure of the concepts themselves. Minimal gaps in Groups 1–2 would indicate that logical composition behaves comparably to semantic composition when the joint configuration is already supported or when the factors remain sufficiently separable. Larger gaps in Groups 3–4 would indicate that logical composition departs from semantic composition most strongly when those structural conditions fail.

The reachability analysis sharpens that interpretation. If SD-IPC reruns remain close to PoE in the easier regimes but fail to recover PoE trajectories in the harder regimes, then the mismatch cannot be attributed only to poor prompt wording. Instead, the expected result would be that some logically composed targets sit outside the region that standard text-conditioned generation reproduces reliably, even when the recovered conditioning signal is derived directly from the PoE image.

This Phase 1 discussion remains deliberately bounded. It does not attempt to resolve whether the underlying cause is representational, algorithmic, or both, nor does it introduce corrections from later phases of the research programme. Its narrower contribution is to frame the composability gap as a structured empirical phenomenon whose severity can be predicted from the relationship between the constituent concepts.

6 Conclusion

Phase 1 formulates semantic composition and logical composition as distinct mechanisms in latent diffusion models and proposes a controlled way to compare them. The central expectation is that the

composability gap is structured: it should remain small for co-occurring or approximately factorized concept pairs, widen for partially overlapping pairs, and become largest for coherent collisions.

The anticipated contribution of the trajectory analysis is to show that this gap is visible in latent dynamics rather than only in decoded outputs. The anticipated contribution of the reachability analysis is to show that SD-IPC-based conditioning does not fully recover the PoE endpoint in the hardest regimes. Taken together, these expected findings motivate Phase 1 as the empirical foundation for later work on why the gap arises and how it might eventually be reduced.

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